Adel Maher Hamdan

Factory Manager

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Professional Summary

Results-driven Factory Manager and Materials Engineer with extensive experience in managing manufacturing operations, product development, and process optimization across the chemical, fertilizer, cosmetics, and pet care industries. Skilled in leading cross-functional teams, implementing GMP/ISO/FDA compliance, and driving efficiency, cost reduction, and product quality improvements. Proven expertise in R&D, with several patents, research papers, and articles published or under publication in product development and production process innovation, some of which have been successfully applied in industrial settings. Adept at strategic planning, supply chain management, PMO framework execution, program governance, and cross-departmental coordination to align production with business objectives. Recognized for strong leadership, innovation, and the ability to deliver sustainable operational and financial results in competitive markets.

Professional Experience

Industrial & Operations Consultant
Freelance / Independent, Amman, Jordan
Sep 2025 – Present

Strategic consultant specializing in manufacturing operations, digital transformation, and process optimization across textile, chemical, and personal care industries. Delivering measurable impact in productivity, cost efficiency, quality, and operational performance through integrated operational and digital solutions.

Key Responsibilities:

- Lead large-scale operational excellence and transformation programs, including process re-engineering, Lean Manufacturing deployment, and performance optimization.
- Advise on factory setup, production line design, and process engineering, including feasibility studies and technical evaluations.
- Design and implement PMO frameworks, governance structures, and execution models to ensure disciplined delivery and effective decision-making.
- Drive ERP-enabled digital transformation, aligning production, supply chain, procurement, and quality processes within integrated systems.
- Integrate ISO 9001, GMP, and regulatory requirements into operational and digital environments to ensure compliance and audit readiness.
- Deliver cost optimization and resource efficiency initiatives, improving productivity while reducing operational expenses.
- Develop and deploy KPIs, OEE systems, dashboards, and executive reporting frameworks to enhance visibility and control.
- Lead change management, stakeholder engagement, and cross-functional alignment to ensure successful transformation and adoption.

Selected Consulting Projects

Textile Manufacturing Company (200+ Employees) – Production Optimization & Operational Transformation

- Led full-scale transformation of production operations through workflow re-engineering and Lean Manufacturing implementation.
- Achieved a 22% increase in productivity while reducing total working hours by ~15% and optimizing working days, delivering significant cost savings and efficiency gains.
- Eliminated process bottlenecks, improved line balancing, and enhanced throughput and production stability.
- Optimized manpower and machine utilization, improving capacity planning and reducing downtime.
- Implemented KPIs, OEE tracking systems, and performance dashboards to sustain continuous improvement.

Personal Care Company (250+ Employees, 6 Production Lines) – Enterprise Digital Transformation (ERP)

- Led enterprise-wide digital transformation initiative across 6 production lines, integrating operations, supply chain, and quality within an ERP ecosystem.
- Re-engineered end-to-end business processes (Procurement, Inventory, Production, Quality) to align with digital workflows and industry best practices.
- Embedded ISO 9001 requirements into the ERP system, enabling real-time compliance, traceability, and audit readiness.
- Developed real-time dashboards, KPI frameworks, and reporting systems, significantly improving visibility and decision-making speed.
- Reduced manual processes and enhanced data accuracy, system integration, and cross-functional coordination.
- Led change management and full-scale user adoption across 250+ employees, ensuring successful transformation and sustained utilization.

Factory Manager

Quatro-P4 Company for Industrial and Commercial Investments, Amman, Jordan

March 2024 – Sep. 2025

Key Responsibilities:

Operations Management:

- Directed end-to-end factory operations to ensure optimal productivity, efficiency, and product quality in the manufacturing of cosmetics and pet care products.

Production Planning & Execution:

- Oversaw production schedules, resource allocation, and process flows to meet daily, monthly, and annual output targets while maintaining GMP and ISO compliance.
- Owned end-to-end program planning, including integrated master plans, resource planning, and dependency management to ensure seamless production continuity.

PMO Governance & Frameworks:

- Led the design, setup, and enhancement of PMO frameworks, governance models, and delivery methodologies across factory operations and improvement programs.
- Drove governance structures, ensuring effective decision-making through steering committees and executive forums aligned with operational objectives.

- Ensured adherence to governance structures including stage gates, approvals, and reporting protocols for all production and improvement initiatives.

Regulatory Compliance & Quality Assurance:

- Ensured adherence to regulatory standards (GMP, ISO, FDA, etc.) across all production and packaging lines; collaborated closely with QA/QC teams to uphold product safety and quality.

Cross-Departmental Leadership:

- Coordinated with R&D, Procurement, Marketing, and Logistics teams to align production with business goals and market demands, supporting both existing products and new product launches.
- Acted as a key point of contact for senior stakeholders, ensuring alignment, transparency, and effective communication across all departments and executive levels.
- Facilitated and led workshops, program reviews, and executive meetings with clear outcomes and actions; maintained clear documentation and follow-up mechanisms.

Performance Reporting & Insights:

- Oversaw the development of high-quality executive reports, dashboards, and program insights to support informed decision-making at the leadership level.
- Monitored program performance, KPIs, and benefits realization, providing actionable recommendations for continuous operational improvement.

Inventory & Supply Chain Management:

- Monitored raw materials and packaging supplies to prevent shortages or overstocking; streamlined procurement processes and vendor communications.

Cost Optimization & Budgeting:

- Implemented process improvements and waste reduction initiatives, contributing to significant cost savings; supported the development and monitoring of departmental budgets.

Risk & Change Management:

- Proactively identified and managed risks, issues, and interdependencies, ensuring timely mitigation and escalation across all operational programs.
- Supported change management and communication efforts across programs, leading the factory through operational challenges, audits, and market fluctuations with minimal disruption.

Team Leadership & Development:

- Led and mentored a multidisciplinary team including production supervisors, engineers, and operators; established KPIs and conducted performance evaluations.
- Guided and mentored junior team members and PMO coordinators in delivery best practices and continuous improvement methodologies.

Health, Safety & Maintenance:

- Enforced workplace safety standards and preventive maintenance programs to minimize downtime and ensure employee well-being.

Business Development Support:

- Contributed to client engagements and business development activities, including proposal development and client presentations, delivering structured analysis and recommendations.

R&D and Quality Senior Officer

Manaseer Fertilizers & Chemicals, Amman, Jordan

March 2023 – March 2024

Key Responsibilities:

R&D Leadership:

- Led and directed R&D teams to develop innovative chemical fertilizers that align with sustainable agricultural needs.
- Developed and formulated equations for powder, liquid, and suspended fertilizers, in addition to inventing gel fertilizers and granulated fertilizers.
- Guided and mentored junior R&D team members, fostering a culture of innovation and structured delivery within the department.

Quality Assurance & Compliance:

- Managed and optimized production processes to ensure compliance with quality standards, including conducting precise tests to verify the efficacy and environmental safety of fertilizers.
- Implemented quality assurance strategies in line with international standards (ISO 9001) to ensure impurity-free products and achieve the highest quality.
- Utilized chemical analysis tools to ensure that fertilizers meet performance standards in both agricultural and environmental criteria.

Project Planning & Governance:

- Supported the establishment and execution of PMO frameworks, standards, and methodologies across R&D and quality improvement projects.
- Developed and maintained integrated project plans, timelines, and deliverables tracking for ongoing R&D initiatives and product development programs.
- Ensured adherence to governance structures including stage gates, approvals, and reporting protocols across all R&D and quality projects.

Stakeholder Coordination & Reporting:

- Collaborated with production teams to integrate innovations into the manufacturing process, ensuring high production efficiency.
- Coordinated with cross-functional stakeholders including production, procurement, and regulatory teams to ensure alignment on R&D objectives and deliverables.
- Prepared and delivered high-quality status reports, dashboards, and management-level presentations on R&D progress, quality KPIs, and product development outcomes.

Risk, Performance & Continuous Improvement:

- Identified, assessed, and monitored project risks, issues, and dependencies within R&D programs; drove mitigation and resolution plans in coordination with relevant teams.
- Monitored project and quality performance against KPIs and provided insights and recommendations for continuous improvement.
- Facilitated technical meetings, workshops, and cross-functional reviews, ensuring clear documentation, action follow-ups, and alignment on program objectives.
- Monitored and applied the latest trends in the fertilizer industry to reduce costs and promote environmental sustainability.

Process Engineer

Manaseer Group Company, Amman, Jordan

December 2020 – March 2023

Key Responsibilities:

- Designed, analyzed, and optimized manufacturing processes to maximize efficiency and minimize costs.

- Conducted feasibility studies and technical assessments for new projects to enhance productivity and product quality.
- Developed engineering solutions to improve operational performance, increase production capacity, and reduce waste.
- Monitored and analyzed operational data to ensure processes meet defined standards and production goals.
- Coordinated with production and maintenance teams to implement technical modifications and resolve process-related issues.
- Ensured the implementation of quality systems and compliance with safety and occupational health regulations.
- Applied advanced data analysis techniques to optimize processes and reduce operational costs.
- Prepared detailed reports for senior management on operational performance and provided actionable recommendations for continuous improvement.
- Supported the establishment and execution of PMO frameworks, standards, and methodologies across engineering projects and improvement programs.
- Developed and maintained integrated project plans, timelines, and deliverables tracking to ensure on-time delivery of process improvement initiatives.
- Identified, assessed, and monitored project risks, issues, and dependencies; drove mitigation and resolution plans in coordination with relevant teams.
- Facilitated meetings, workshops, and cross-functional reviews, ensuring clear documentation, follow-ups, and alignment on project objectives and deliverables.
- Prepared and delivered high-quality status reports, dashboards, and presentations for senior management and engineering leadership.
- Monitored project performance against KPIs and provided insights for continuous improvement across operational and engineering programs.

Quality Engineer (QA / QC)
Al-Moasron Group Company, Amman, Jordan
August 2020 – December 2020

Key Responsibilities:

- Developed and implemented quality management systems (ISO 9001) to ensure plastic products comply with international and local standards.
- Conducted thorough testing and inspections on raw materials and final products to ensure their quality and safety meet specifications.
- Monitored and analyzed plastic manufacturing processes to identify potential improvements and achieve high production efficiency.
- Coordinated with production teams to ensure the application of quality procedures throughout all stages of production, from raw materials to finished products.
- Developed regular technical reports on quality performance and provided recommendations for process improvement and waste reduction.
- Utilized Six Sigma and Lean Manufacturing techniques to improve production processes and reduce defects in plastic products.
- Participated in training teams on best practices in quality assurance and root cause analysis to resolve quality issues.
- Managed customer complaints and provided appropriate solutions to improve customer satisfaction and ensure their requirements were met.

Education

Balqa'a Applied University
Bachelor's Degree in Materials Engineering | 2015 – 2020 | GPA: Good
Graduated Project: Copper extraction from Jordanian ores

Publication

Athamneh, T., Al-Najjar, M.A.A., Garafat, R., Abuawad, A.M., Alshweiat, A., Barakat, M., AbuIrmaileh, W.F., Hamdan, A.M., Odat, T.A., Altarabeen, R., et al. (2025). Preparation and Characterization of Copper-Crosslinked Alginate–Hyaluronic Acid Aerogels as Potential Wound Dressing Materials with Enhanced Antibacterial Properties. *Polymers, 17, 2406.* https://doi.org/10.3390/polym17172406

Certifications and Training

- NDT / RT welding inspection, Jordan Engineering Association, April 2017
- ISO 9001 Quality Management Training, ZRIQAT Engineering Academy, Oct. 2017
- NDT/MT&UT&PT welding inspection, Jordan Engineering Association, Sep. 2018
- cGMP (Good Manufacturing Practice), Jordan Engineering Association, Oct. 2018
- Safety and Occupational Health, Jordan Engineering Association, May 2019
- Occupational Health & Safety Training, UDEMY, April 2022
- NEBOSH International H&S Management System, UDEMY, April 2022
- OSHA Workplace Safety, UDEMY, April 2022

Technical Skills

- Manufacturing Process Optimization – design, analysis, and improvement of chemical, cosmetics, pet care, and fertilizer production lines.
- Production Planning & Execution – scheduling, resource allocation, workflow optimization, and integrated master planning.
- PMO Frameworks & Governance – setup and execution of PMO standards, methodologies, stage-gate governance, and executive reporting.
- Quality Assurance & Control – GMP, ISO 9001, FDA compliance, Six Sigma, Lean Manufacturing, and root cause analysis.
- Chemical Formulation & R&D – development of fertilizers (powder, liquid, suspension, gel, granulated) and cosmetic/pet care formulations.
- Process Engineering – feasibility studies, technical assessments, process simulations, and continuous improvement.
- Risk & Change Management – risk identification, mitigation planning, dependency management, and change communication.
- Regulatory Compliance – ensuring alignment with local and international standards (ISO, GMP, FDA, HSE).
- Data Analysis & Reporting – operational performance tracking, KPI monitoring, dashboards, and executive-level presentations.
- Testing & Validation – material testing, chemical analysis, and performance verification.
- Health, Safety & Environment (HSE) – safety audits, preventive maintenance, and environmental sustainability strategies.

- Supply Chain & Inventory Management – procurement, vendor management, and raw material optimization.

Administrative Skills

- Leadership & Team Development – leading cross-functional teams, mentoring engineers, supervisors, and junior PMO team members; performance evaluations and training programs.
- Strategic Management – supporting executive decisions, crisis management, and long-term operational planning.
- Financial Management – budgeting, cost reduction, waste minimization, and ROI-driven resource allocation.
- PMO Framework Design & Execution – establishing and enhancing PMO standards, governance models, delivery methodologies, and stage-gate compliance across programs.
- Program & Project Management – end-to-end program planning, integrated master plans, resource planning, dependency management, and deliverables tracking across factory, R&D, and engineering programs.
- Governance & Decision Support – driving governance structures, steering committees, executive forums, and approvals to enable timely, effective decision-making.
- Risk & Issue Management – proactive identification, assessment, and mitigation of project risks, issues, and interdependencies with timely escalation and resolution.
- Executive Reporting & Dashboards – overseeing the development of high-quality status reports, dashboards, and program insights for senior leadership and stakeholders.
- Stakeholder Management – acting as key point of contact for senior stakeholders; ensuring alignment, transparency, and effective communication across multi-stream programs.
- Facilitation & Workshop Leadership – facilitating and leading workshops, program reviews, steering committees, and executive meetings with clear outcomes and documented follow-ups.
- Change Management & Communication – leading programs through transformation, audits, market fluctuations, and operational challenges with structured communication plans.
- Cross-Departmental Coordination – liaising between R&D, procurement, marketing, logistics, and finance for business alignment and program delivery.
- Business Development Support – contributing to client engagements, proposal development, structured analysis, and executive presentations.
- Employee Engagement & HR Support – establishing KPIs, promoting safety culture, and supporting recruitment and training initiatives.

Projects & Research Initiatives

Materials Engineering & Process Development:

- Copper Extraction from Jordanian Ore
- Preparation of Dead Burned Magnesia using Dolime
- Preparation of Dead Burned Magnesia using Lime
- Thermoplastic Powder Coating
- Composite Materials of Basalt Powder and Polyethylene
- Composite Materials of Basalt Powder and Polypropylene
- Improvement of Ground Calcium Carbonate Brightness
- Kaolin Particles Coated with Ground Calcium Carbonate
- Replacement of Titanium Dioxide Pigment

- Developing a Process for Preparing Magnesia Using a Continuous Cyclic Process
- Kaolin Beneficiation
- PVC Pipe Formulation Development
- Silica Sand Washing
- Surface-Reacted Calcium Carbonate
- Surface-Reacted Calcium Carbonate with Bleaching
- Process of Utilizing Limestone Saw Mud in Cement Production
- Phosphate-Based Cement
- Developing the Fluidized Bed Process from Scratch
- Sand Blasting Development Process
- Pozzolana Beneficiation Process
- Method for Simultaneous Production of Magnesium Derivatives and Lithium Carbonate from High Concentration Magnesium Brine

Fertilizer Research & Development:

- Jordanian Zeolite Ore for Agriculture and Animal Uses
- Jordanian Basalt Ore for Agricultural Uses
- Dispersion of Acidic-Based Liquid Fertilizers P-K
- Development of Natural Fertilizers
- Controlled-Release Natural-Based Fertilizers
- Development of N-P-K Fertilizers
- Development of Nano Fertilizers
- Improvement of Suspension Fertilizers
- Development of Hydrogel Fertilizers
- Enhancing Plant Immunity and Agricultural Productivity Using Nano-Biological Systems to Support Food Security

Industrial & Business Development:

- Establishing, Launching, and Managing a Manufacturing Company to Produce Pet Care and Personal Care Products from Scratch to Export

Languages

- Arabic (Native)
- English (Good)

References

Available upon request.